Sub-sampling PLL with feedforward noise cancellation

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A 1.6 GHz ring-oscillator PLL for SKY130 that multiplies a 50 MHz crystal reference by 32 and divides the result into 800, 400, 200 and 100 MHz clocks. A sub-sampling phase detector keeps detector noise out of the N^2 multiplication, and feedforward noise cancellation removes about a third of the remaining oscillator jitter. It runs from the harness 3.3 V analog and 1.8 V digital supplies, one reference current and the reference voltage. Target: 1.4 ps rms jitter (10 kHz–40 MHz) at about 9 mW in under 0.12 mm^2, to be verified post-layout with open-source tools.

1. Type of IP block

An integer-N clock-multiplying PLL (phase-locked loop) for system-on-chip clocking, with a fixed multiplication ratio N = 32. An external 45–55 MHz crystal oscillator gives 1.44–1.76 GHz. The nominal 50 MHz reference gives 1.600 GHz, which on-chip binary dividers turn into 800, 400, 200 and 100 MHz. The ÷2 output (800 MHz, with an exact 50% duty cycle) is the main system clock; the full-rate 1.6 GHz clock is also available for designs that need it. The architecture is a ring-oscillator sub-sampling PLL with feedforward noise cancellation, after Nagam and Kinget (IEEE JSSC, 2018). The high-speed signal path uses 1.8 V core devices. Everything that connects to the 3.3 V supply or to a pad uses 3.3 V-rated devices.

The first use is a low-jitter sampling clock for data converters: 1.4 ps rms keeps a 100 MHz input at about 61 dB SNR, roughly 10 bits. The divided outputs also give SKY130 digital logic and I/O clean 100–400 MHz clocks. No PLL was among the 25 blocks taped out in Chipalooza Challenge #1, whose oscillators are all low-frequency, and this block fills that gap.

2. I/O list, including test ports

On the Chipalooza harness the block uses all three dedicated pads, all four shared analog lines (for DC and low-frequency signals only), the bias voltage, one reference current, 21 of the 24 digital controls and 5 of the 12 digital test outputs. Every pad-facing node connects only to 3.3 V-rated devices, so any pad voltage from 0 V to VDDA leaves the 1.8 V core unstressed. All pads are standard SKY130 I/O library cells; no custom pad is needed.

Signal	Type	Connection	Function
REF_IN	Analog input	Dedicated pad 1, analog pad. Needs a direct, low-resistance path to the pad	45–55 MHz reference from an external crystal oscillator. On-chip AC coupling and a self-biased 3.3 V receiver: the pad sits at VDDA/2 and accepts anything from 0.4 V pp to a full 0 V to VDDA square wave
CLK_OUT	Analog output	Dedicated pad 2, analog pad. Needs a direct, low-resistance path to the pad	Measurement clock: f_out/16 (100 MHz nominal), f_out/32 or f_out/64, selected by clkout_sel. Open-drain 3.3 V driver sinking 4 mA; the test board provides 50 Ω to VDDA, or a bias tee, into the instrument
CLK_DIG	Digital output	Dedicated pad 3, standard digital pad (1.8 V core side)	f_out/64 (25 MHz nominal) CMOS copy for a frequency counter or oscilloscope, inside the digital pad's speed range
ANA0	Analog input	Shared analog line 0	VTEST: a DC level or a tone of 1 MHz or less injected into the oscillator current feed, for cancellation calibration and supply-sensitivity tests. Disconnected when vtest_en = 0
ANA1–ANA3	Analog outputs	Shared analog lines 1–3	Buffered DC monitors: tuning voltage; ring supply node; and, selected by amon_sel, the regulator output, the held phase-detector voltage or a bias test current. High impedance when amon_en = 0
VDDA	Power	Slot 3.3 V	Oscillator current DAC, regulator, reference receiver, output driver, monitors

Signal	Type	Connection	Function
VDDD	Power	Slot 1.8 V	Dividers, frequency-loop logic, control registers, CLK_DIG path
VSSA, VSSD	Ground	Slot grounds	Returns of the two supplies
VREF	Bias input	Bandgap-referenced bias voltage, set to 1.2 V	Regulator reference
IBIAS	Bias input	1 bandgap-referenced current source, set to 5 µA	Master bias for the charge pump, buffers, current DAC and regulator

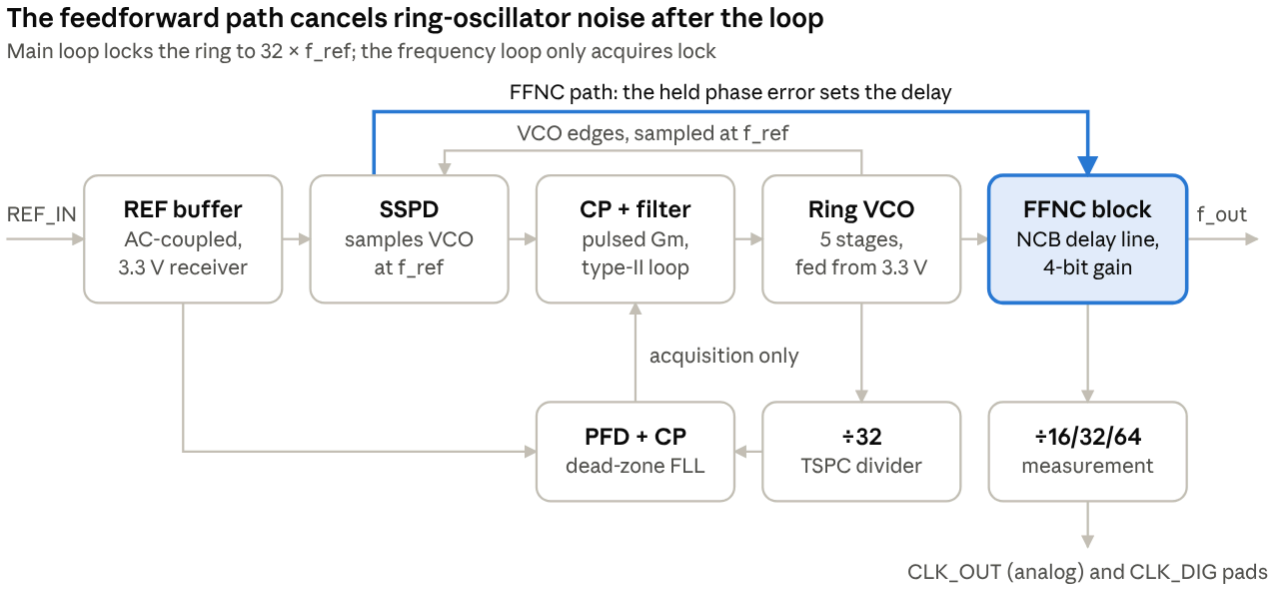
The reference current flows into IBIAS. If the harness specification fixes the opposite direction, only the input mirror changes polarity in layout; the specification does not change.

Digital controls (21 of 24): pll_en, fll_en, ffnc_en, cco_code[4:0] (coarse frequency), ncb_code[3:0] (cancellation gain), icp_trim[1:0], clkout_sel[1:0] (/16, /32, /64 or off), clkdig_en, amon_sel[1:0], amon_en, vtest_en. All default to 0, which leaves the block off and its pads quiet. Three are reserved.

Digital test outputs (5 of 12): lock_det, fll_active, fll_dir (too fast or too slow), cal_out (sign of the calibration error), ldo_ok (regulator in regulation).

Core-level outputs for reuse in other chips: CLK_FULL (f_out) and CLK_DIV2 to CLK_DIV16. They are not brought out on the harness.

3. Functional description



Block diagram · main loop, feedforward cancellation, frequency acquisition, measurement output

The loop locks a 5-stage pseudo-differential ring oscillator to 32 × f_ref. The noise-cancellation block then removes most of the ring's remaining jitter before the clock leaves the block.

- **Sub-sampling phase detector (SSPD).** It samples the 1.6 GHz oscillator output directly at the reference rate, and the held voltage is the phase error. The main loop has no feedback divider, so phase-detector and charge-pump noise are not multiplied by N^2.
- **Charge pump and loop filter.** A pulsed-Gm charge pump and a type-II RC filter set the fine tuning voltage, with a loop bandwidth of about 2.5 MHz.
- **Ring oscillator.** Five pseudo-differential stages. The odd stage count makes the common-mode loop inverting, so the ring cannot latch at power-up. A 5-bit current DAC built from 3.3 V devices feeds the ring from VDDA and sets the coarse frequency. Its range and an on-chip clamp keep the ring's supply node below 1.8 V at every code and corner.
- **Feedforward noise cancellation (FFNC), implemented by the noise-cancellation block (NCB).** The held phase error also drives a voltage-controlled delay line after the oscillator. With its gain set to the inverse of the detector gain, the delay subtracts the oscillator's own phase deviation, up to about f_ref/8 (6 MHz). A 4-bit code sets the gain and is calibrated in the foreground with a test tone. With the block disabled, the PLL works normally without cancellation.
- **Frequency acquisition.** A sub-sampling detector locks to whichever harmonic is nearest, so a dead-zone frequency-locked loop (÷32 divider, PFD, charge pump) pulls the oscillator to 32 × f_ref. It goes quiet once the phase error is inside its dead zone,

and catches any slip to a neighbouring harmonic.

- **Supplies and bias.** An on-chip low-dropout regulator built from 3.3 V devices makes a clean 1.8 V rail from VDDA for the phase detector, charge pump, oscillator buffers and delay line. It is referenced to the harness bias voltage. The dividers, frequency-loop logic and control registers run on the 1.8 V digital supply. The reference current sets every bias current through mirrors.
- **Level shifters.** They sit only where a signal crosses between 1.8 V and 3.3 V: the controls of the 3.3 V circuits (current-DAC code, regulator, output-driver and monitor enables), the measurement clock into the 3.3 V output driver, and the reference receiver into the 1.8 V core. The frequency detector (1.8 V digital) drives its charge pump (1.8 V regulated) through a same-voltage domain buffer, so no level shifter is needed there.
- **Outputs.** The full-rate clock and its ÷2, ÷4, ÷8 and ÷16 taps (800, 400, 200 and 100 MHz at a 50 MHz reference) go to the system-on-chip. For measurement, a ÷16, ÷32 or ÷64 copy drives an open-drain 3.3 V driver on CLK_OUT, and a ÷64 copy (25 MHz) leaves on a standard digital pad as CLK_DIG.

4. Target specification

All values are design targets. Minimum and maximum values are to be met across the full operating range below in post-layout verification over process, voltage and temperature. Typical values are for the tt corner at 27 °C, nominal supplies and bias, a 50 MHz reference and calibrated codes. Unless noted, performance is at the oscillator output with noise cancellation (FFNC) on. The maximum values keep about twice the margin of the current pre-layout estimates, so the specification should not need to change during the design.

Operating conditions

Parameter	Condition	Min	Typ	Max	Unit
VDDA	Slot analog supply	2.97	3.3	3.63	V
VDDD	Slot digital supply	1.71	1.8	1.89	V
VREF	Bandgap-referenced bias voltage	1.14	1.20	1.26	V
IBIAS	Bandgap-referenced current	4.0	5.0	6.0	µA
Temperature	Junction	−40	27	110	°C
Reference frequency f_ref	External crystal oscillator	45	50	55	MHz
Reference amplitude	Sine or square at REF_IN	0.4	1.0	3.3	V pp
Reference phase-noise floor	100 kHz offset and above; condition for the jitter and phase-noise rows			−155	dBc/Hz

Parameter	Condition	Min	Typ	Max	Unit
CLK_OUT termination	Board: 50 Ω to VDDA or a bias tee, into a 50 Ω instrument		50		Ω

Performance

Parameter	Condition	Min	Typ	Max	Unit
Output frequency f_out	32 × f_ref	1.440	1.600	1.760	GHz
Divided outputs (core level)	f_out/2, /4, /8, /16 at f_ref = 50 MHz		800, 400, 200, 100		MHz
CLK_OUT frequency	/16 setting	90	100	110	MHz
CLK_OUT sink current	Driver enabled	3	4	6	mA
CLK_DIG frequency	f_out/64	22.5	25	27.5	MHz
RMS jitter, FFNC off	Integrated 10 kHz–40 MHz		2.3	4.7	ps
RMS jitter, FFNC on	Integrated 10 kHz–40 MHz		1.4	3.2	ps
Jitter reduction by FFNC	After gain calibration	20	35		%
Phase noise, FFNC off	300 kHz offset		−110	−102	dBc/Hz
Phase noise, FFNC on	300 kHz offset		−120	−110	dBc/Hz
Reference spur, FFNC off	At f_out ± f_ref		−55	−45	dBc
Reference spur, FFNC on	At f_out ± f_ref		−50	−42	dBc
Lock time	From pll_en with coarse code set, until the phase error stays within ±10 ps		5	20	µs
Regulated analog rail	Internal, at ANA3	1.71	1.80	1.89	V
Power, FFNC on	All supplies; PLL core including the regulator; excludes output taps and pad drivers		9.0	15	mW
Power, FFNC off	Noise-cancellation block disabled		7.5	13	mW
Power, output dividers	All four taps (/2 to /16) enabled, from VDDD		0.5	1.0	mW
Jitter figure of merit	$10 \cdot \log10[(\sigma/1\,\mathrm{s})^2 \cdot (P/1\,\mathrm{mW})]$, FFNC on		−227.5		dB
Area	Standalone layout, excluding pads		0.08	0.12	mm²

Absolute limits (exceeding these may damage the block; operation is only guaranteed within the operating conditions)

Parameter	Min	Max	Unit
VDDA	0	3.96	V
VDDD	0	1.95	V
REF_IN, CLK_OUT, ANA0–ANA3, VREF, IBIAS pin voltage	−0.3	VDDA + 0.3	V
CLK_DIG	As the standard digital pad		
Internal ring supply node, any cco_code		1.80	V
Internal regulated rail		1.95	V
Junction temperature	−40	125	°C

The two internal limits are design constraints, not user settings: the current-DAC range, an on-chip clamp and the regulator keep every 1.8 V device within its rating at all codes and corners.

How the specification will be verified. Every value above will be verified post-layout with open-source tools only: xschem, ngspice, Magic, netgen, KLayout and CACE in the IIC-OSIC-TOOLS container. The corners are tt, ss, ff, sf and fs; VDDA ±10%, VDDD ±5%, VREF ±5% and IBIAS ±20%; −40, 27 and 110 °C; and reference frequencies of 45, 50 and 55 MHz, with Monte Carlo mismatch on the phase detector, charge pump, noise-cancellation block, regulator and current DAC. Jitter is built from ngspice simulations: oscillator phase noise from its impulse sensitivity function, the other blocks' noise from noise analysis, combined in a Python phase-domain model. This method overestimates oscillator phase noise, so a pass with it is a pass with margin. The schematics, layout, testbenches, scripts and final DRC and LVS reports will be in a public repository under the Apache 2.0 license.

5. Test plan outline

Every measurement uses only the three dedicated pads, the shared analog lines and the digital controls; the 1.6 GHz clock never leaves the chip. Cancellation on and off are compared on the same chip, pad, cable and instrument, so the jitter reduction is measured reliably even close to the instrument's noise floor.

1. **Bring-up and power.** Apply VDDA and VDDD, set VREF to 1.2 V and IBIAS to 5 µA, and check SPI access and ldo_ok. Read the regulator output and ring supply node on the analog monitors. Measure the VDDA and VDDD slot currents with pll_en, ffnc_en and each output enabled in turn, which gives the power of each part.

2. **Coarse calibration.** With the frequency loop enabled, sweep cco_code and record the codes where lock_det goes high. Repeat at each test temperature.
3. **Frequency and range.** Measure CLK_DIG on a frequency counter: 25.000 MHz at f_ref = 50 MHz confirms N = 32. Repeat at f_ref = 45 and 55 MHz to confirm the range.
4. **Lock time.** Put REF_IN and CLK_DIG on an oscilloscope, triggered by the SPI write that sets pll_en. Track the REF_IN-to-CLK_DIG edge delay with the time-interval function until it stays within ±10 ps.
5. **Phase noise and jitter.** Measure CLK_OUT at /16 (100 MHz) on a phase-noise analyzer, or a spectrum analyzer with a phase-noise option. Add 20·log10(16) = 24.1 dB to refer phase noise to the oscillator output. Report rms jitter integrated from 10 kHz to 40 MHz directly: absolute jitter in seconds passes through an ideal divider unchanged. Repeat with ffnc_en off and on.
6. **Cancellation calibration.** Inject a 50 kHz tone on ANA0 (VTEST) and step ncb_code from 0 to 15. Keep the code with the smallest tone sideband at CLK_OUT, then confirm the jitter reduction against step 5.
7. **Reference spur.** With N = 32, the spur sidebands on the 100 MHz output fall at 50 and 150 MHz, on top of the reference frequency and its third harmonic. Measure both lines in lock, then again with pll_en off and the reference still running, which shows the reference feedthrough alone. A true spur is a phase-modulation sideband, so the two lines are equal in level and stand well above the feedthrough. Add 24.1 dB to refer the spur to the oscillator output.
8. **Supply-noise sensitivity.** Inject 1 mV pp at 100 kHz on ANA0 and compare the resulting spur with cancellation on and off.
9. **Temperature.** Repeat steps 3 to 7 at the coldest and hottest temperatures the available equipment allows, up to −40 and 110 °C.
10. **Report.** Compare every result with the specification in section 4 and publish the measured results, plots and the scripts that produced them in the repository.

Equipment needed: a test board with low-noise 45, 50 and 55 MHz crystal oscillators (phase-noise floor of −155 dBc/Hz or lower) and the 50 Ω CLK_OUT termination; a phase-noise analyzer, or a spectrum analyzer to at least 200 MHz with a phase-noise option; a 1 GHz oscilloscope with a time-interval function; a function generator for the test tone; a frequency counter; bench supplies or a source-measure unit with current readout; and an SPI host. Measurement will take place at an external lab facility, listed with its equipment in the separate attachment.